

Spatio-temporal evaluation of land use and cover change and prediction of habitat quality in Estuarine Delta National Parks: a case study of the Yellow River Estuary National Park

Zhuang Zhuang · Zhaojin Li · Pengfei Tai · Fugao Jiang

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Abstract

As a critical component of China's national park system, the Yellow River Estuary National Park exhibits a unique habitat evolution mechanism shaped by the interplay between anthropogenic disturbances and ecological restoration within a land–sea interaction zone. This study simulates land use and land cover change and assesses habitat quality dynamics from 2003 to 2023, with scenario projections for 2033, by integrating the InVEST and PLUS models. The results reveal a dual-process mechanism underlying habitat change: urbanization drives the expansion of construction land, which encroaches upon cropland and barren land, leading to widespread habitat degradation and fragmentation; concurrently, wetland restoration and water body expansion, driven by ecological policies, partially offset these losses by locally enhancing habitat quality in coastal zones. This coexistence of degradation and restoration shapes a spatially heterogeneous pattern, where low-quality habitat expands and consolidates around urban agglomerations, while high-quality habitat becomes increasingly fragmented and confined to remnant wetland patches. Scenario projections indicate a continuation of these trends, with further expansion of low-quality habitat and sustained fragmentation of high-quality areas, suggesting limited ecological resilience under current land use trajectories. By uncovering the process-based linkages between land use transformation and habitat quality response, this study provides a mechanistic understanding of estuarine habitat evolution and offers a scientific foundation for spatial planning and ecological governance in integrated land–sea national parks. These insights contribute to broader discourses on land quality and ecosystem service sustainability in deltaic social-ecological systems.

Keywords Yellow River Estuary National Park · Habitat quality · Land use and cover change · InVEST model · PLUS model · Ecosystem services

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Introduction

The establishment of a nature reserve system with national parks as the main body in China is an important institutional innovation in the construction of global ecological civilization construction and a key practice to realize the harmonious coexistence between man and nature¹. At present, China has established a national park system to protect the authenticity and integrity of the natural ecosystem and build a national ecological security barrier^{2,3}. Yellow River Estuary National Park is the first Marine land integrated national park in China. It undertakes the important mission of ecological protection and high-quality development in the Yellow River Basin⁴. It is an important practice to fully implement the concept that “lucid waters and lush mountains are invaluable assets”, adhere to harmonious coexistence between man and nature, and promote all-round, all-round, and all-round ecological protection^{5,6}. As a key indicator of ecosystem health, habitat quality directly reflects the capacity of landscapes to provide essential ecosystem services, such as biodiversity maintenance, water purification, and climate regulation. Moreover, land use changes not only alter habitat patterns but also affect land quality which in turn influences the delivery of these services. Therefore, assessing habitat quality dynamics within the framework of land quality and ecosystem services is critical for guiding national park management and regional sustainability. In 2021, the Chinese government formulated the outline of the Yellow River Basin ecological protection and high-quality development plan, indicating that the high-quality ecological protection and high-quality development of Yellow River Estuary National Park have been incorporated into the national strategic plan.

Although previous studies have extensively examined the spatial and temporal variation of habitat quality under land-use change, most have emphasized changes in habitat quality patterns and future trends. Less attention has been paid to the relationship between specific land-use transitions and their corresponding habitat quality responses. In estuarine wetlands, where human activities and ecological restoration jointly influence land-use change, overall habitat quality trends may conceal spatially differentiated processes of degradation and improvement. Clarifying this relationship is therefore important for understanding habitat quality evolution in estuarine protected areas.

Against this background, this study examines the spatio-temporal relationship between land-use change and habitat quality in the Yellow River Estuary National Park. The PLUS model is employed to characterize and simulate land-use dynamics, while the InVEST model is used to evaluate habitat quality under observed and projected land-use patterns. (Fig. 1). At the level of land use simulation, the multi-factor driven land use change simulation prediction is constructed based on the PLUS model to identify the leading driving mechanism of land use spatial evolution in the study area. The land use pattern in 2023 was simulated by using the land use data in 2003 and 2013, and compared with the actual land use results in 2023 to evaluate the accuracy and applicability of the model. On the premise that the reliability of the model has been verified, further use the land use change from 2013 to 2023 to predict the Scenario prediction of land use pattern in 2033, providing a spatial basis for habitat quality analysis. Finally, based on the land use data at different stages, the simulation results of the PLUS model are used as an important input of the InVEST model to quantitatively evaluate the temporal and spatial distribution characteristics of habitat quality in the study area in 2003, 2013, 2023 and 2033. The system reveals the potential ecological risk of land use change on habitat quality, so as to realize the comprehensive analysis of the evolution process of habitat quality evolution in Yellow River Estuary National Park.

Accordingly, the central scientific question of this study is how different land-use transitions are associated with spatially differentiated changes in habitat quality within a dynamic estuarine landscape. The research results can not only provide a scientific basis for the space control, ecological restoration and adaptive management of Yellow River Estuary National Park, but also provide a theoretical basis and typical cases for the research and practice of similar land and sea integrated protected areas in the world.

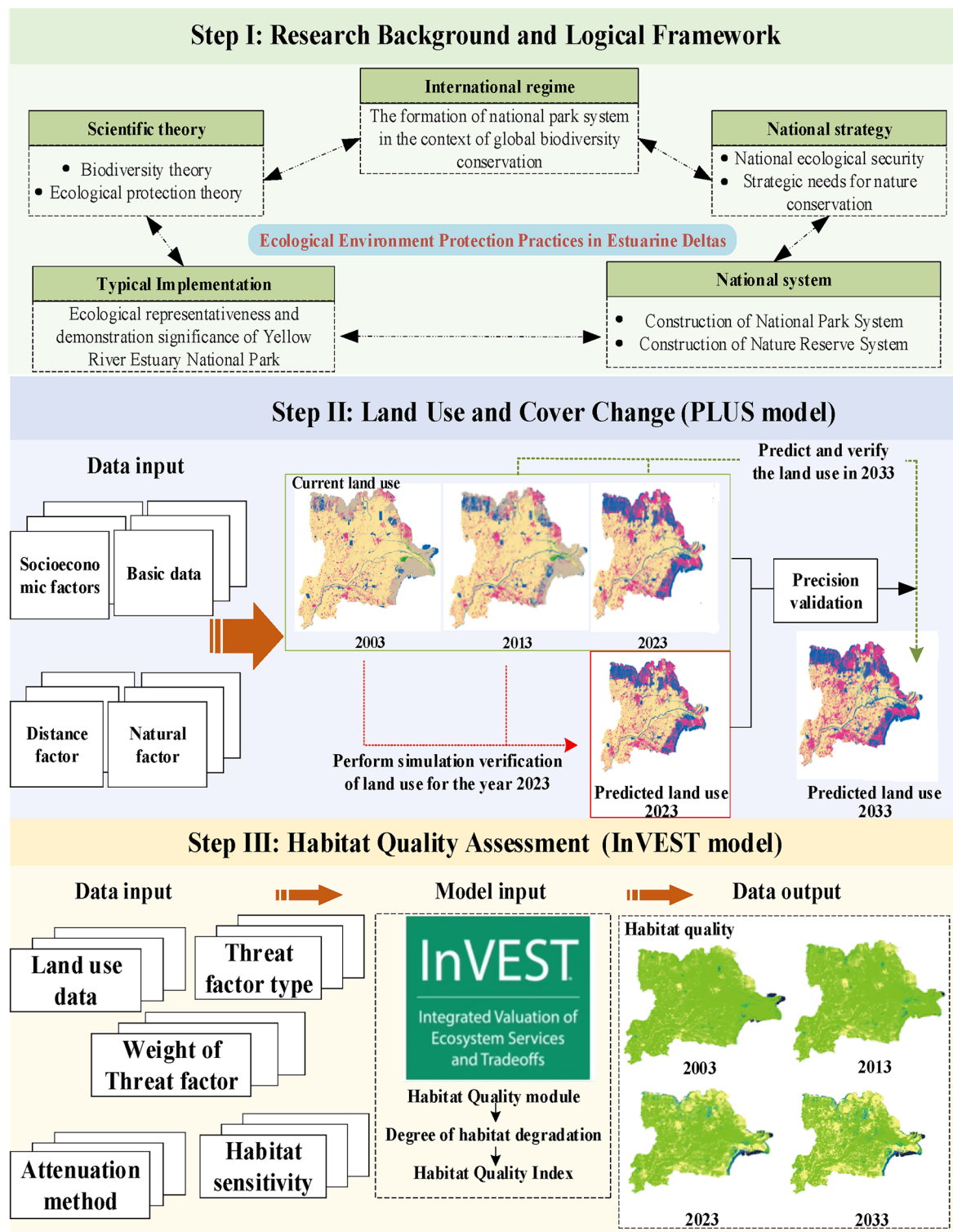


Fig. 1 Research roadmap.

Literature review

Habitat quality is a key indicator to measure ecosystem health and biodiversity sustainability, and its core theory has evolved over time^{7,8}. In the early stage, the definition of habitat quality mainly focused on providing basic living conditions such as food, water and breeding sites for species⁹. With the deepening of ecological research, its connotation extends to the comprehensive assessment of ecological integrity and ecosystem services¹⁰. Ecological integrity emphasizes the integrity and stability of ecosystem structure, function and processes. A healthy ecosystem can resist external interference and maintain its original state for a long time^{11,12}. Ecosystem service functions link habitat quality with human well-being, emphasizing the ability of habitat to provide support services, supply services and cultural services^{13,14}. Based on the evolution of the above theoretical connotation, research on habitat quality has gradually formed a theoretical system with landscape ecology theory and Biodiversity Conservation Theory as the core. Landscape ecology emphasizes the impact of spatial pattern on ecological processes, and believes that Land Use and Land Cover Change (LUCC) is a key factor affecting habitat quality^{15,16}. Rapid urbanization and agricultural expansion will lead to habitat fragmentation and connectivity decline^{17,18}. The theory of ecological diversity protection is the core driving force for the study of habitat quality, and habitat degradation is the main reason for the decline in biodiversity^{19,20}. In summary, habitat quality is understood as a dynamic, multi-scale, complex system affected by both natural and human factors²¹.

The habitat quality assessment method system covers the traditional field survey method and model evaluation method. The traditional field survey method has the advantages of high data authenticity and rich details by directly obtaining the data on species distribution and habitat characteristics²². However, these methods are often expensive and difficult to operate on a large scale or long time series. The RSEI model is based on remote sensing data to rapidly assess the regional ecological environment quality by integrating indicators such as greenness, humidity, heat and rainfall²³. The MaxEnt model is often used to predict species distribution and assess habitat suitability by associating species appearance data with environmental factors²⁴. The CA-Markov model is mainly used to simulate land use change and indirectly assess habitat quality changes^{25,26}. These models have their own advantages, but the common challenge is how to accurately obtain high-quality input data, calibrate model parameters and deal with spatial heterogeneity²⁷. The InVEST model is widely used because it can quantify the degree of habitat threatened and assess habitat quality^{28,29}. The habitat quality index is calculated by the sensitivity of land cover types to threat sources and the impact range of threat sources, which includes multiple factors such as economy, nature, society, and land use that jointly affect habitat quality³⁰. This method breaks through the inherent limitations of traditional field survey and provides the feasibility for high-precision and dynamic habitat quality assessment. Among these approaches, the InVEST model's habitat quality module has been widely applied because it links land use/cover patterns to ecosystem service outcomes. By quantifying the impacts of threats on habitat suitability, it provides a spatially explicit assessment of a key supporting service which underpins biodiversity and other regulating and cultural services. This makes the InVEST model particularly suitable for studies framed within the land quality–ecosystem services nexus. These methods differ in their data requirements and applicability: field surveys provide detailed ecological information but are difficult to apply over large areas, RSEI is suitable for regional environmental assessment, MaxEnt is more appropriate for species-specific habitat suitability analysis, whereas InVEST provides a spatially explicit framework for evaluating habitat quality responses to land-use change.

Land-use simulation has gradually evolved from traditional cellular automata approaches toward machine-learning and deep-learning methods capable of capturing complex spatio-temporal patterns. These approaches differ in their data requirements, interpretability, and analytical purposes. Deep-learning methods offer advantages in learning complex spatial and temporal dependencies from data-rich time series, whereas PLUS is well suited to identifying the driving factors of land expansion and reproducing spatially explicit land-use transitions. Compared with purely prediction-oriented approaches, process-based land-use models such as PLUS place greater emphasis on identifying land expansion rules and their associated driving factors, thereby providing a

more interpretable basis for linking land-use transitions with subsequent ecological responses. In this study, PLUS was selected because its land expansion analysis and patch-generation mechanisms are consistent with the objective of examining land-use transitions and their associated habitat quality responses.

The particularity of estuarine wetland habitat quality is that its rivers and coastlines intersect each other, forming a unique ecological environment. Hydrological connectivity, such as runoff and sediment transport, is crucial to the formation and maintenance of estuarine wetland habitats³¹. Studies have shown that estuarine wetlands are important biodiversity hot-spots and are vulnerable to human activities and climate change³². Therefore, it is particularly important to evaluate and protect the habitat quality of estuarine wetlands. The existing research on the Yellow River Estuary covers wetland evolution in the Yellow River Delta, habitat suitability of rare species, and the impact of human activities^{31–32}. However, relatively limited attention has been paid to the integrated coupling of habitat quality evaluation, simulation, prediction, and the quantification of underlying driving mechanisms^{34,35}. Existing studies have tended to focus on assessments at a single time point or scale, with comparatively less emphasis on comprehensive analyses that integrate historical evolution, current conditions, and future projections^{36,34}. Moreover, although previous studies have examined the effects of human activities, climate change, and other driving factors, how these factors interact and how different land-use transitions are associated with corresponding habitat quality responses in the Yellow River Estuary remain insufficiently understood³².

Therefore, the PLUS model and InVEST model are used to provide effective tools for land use simulation and habitat quality assessment, respectively. The PLUS model can accurately simulate the spatio-temporal evolution of land use pattern³⁷, while the InVEST model can quantitatively assess habitat quality based on land use pattern^{38,39}. The integration of PLUS and InVEST combines the spatial simulation capability of land-use change with habitat quality assessment, enabling historical land-use transitions, future scenarios, and corresponding habitat quality responses to be examined within a unified spatio-temporal framework⁴⁰.

Overview of the study area and data sources

Overview of the study area

Yellow River Estuary National Park (37°25'02"—38°17'53"N, 118°13'55"—119°30'57"E) is the first land-sea integrated national park in China. This study defines its research area based on the scope of three county-level administrative divisions—Kenli District, Hekou District, and Lijin County—as delineated in official documents including “*the Construction Plan of the Yellow River Estuary National Park*” and “*the Boundary and Management Zoning Map of the Yellow River Estuary National Park*” issued by the Shandong Provincial Department of Natural Resources (Shandong Provincial Forestry Bureau) (Fig. 2). The region is located in the monsoon climate zone of the warm zone, with four distinct seasons and obvious interannual variation of precipitation. The land-form is dominated by a delta plain and tidal flat, and the dynamic change of coastline is significant. Yellow River Estuary National Park has the most complete estuarine wetland ecosystem in China’s warm zone and the most active land and sea changes. It is a typical representative of the estuarine wetland ecosystem in the world.

Data source and processing

The data used in the study include land use data⁴¹, socioeconomic factors⁴², distance factor⁴³ and natural factor⁴⁴ (Table 1). The land use data was obtained from the CLCD dataset provided by Wuhan University. To maintain consistency between the PLUS-based land use simulation and the investment-based habitat quality assessment approach, the land cover data for 2003, 2013, and 2023 were categorized into five classes using ArcMap 10.8.1 (<https://www.esri.com>): cropland, grassland, water, barren land, and built-up land. This classification scheme was applied uniformly across all study periods to ensure comparability in analyses of land use transitions and habitat quality changes. Considering the accuracy and practical significance of the PLUS model, 12 driving factors were

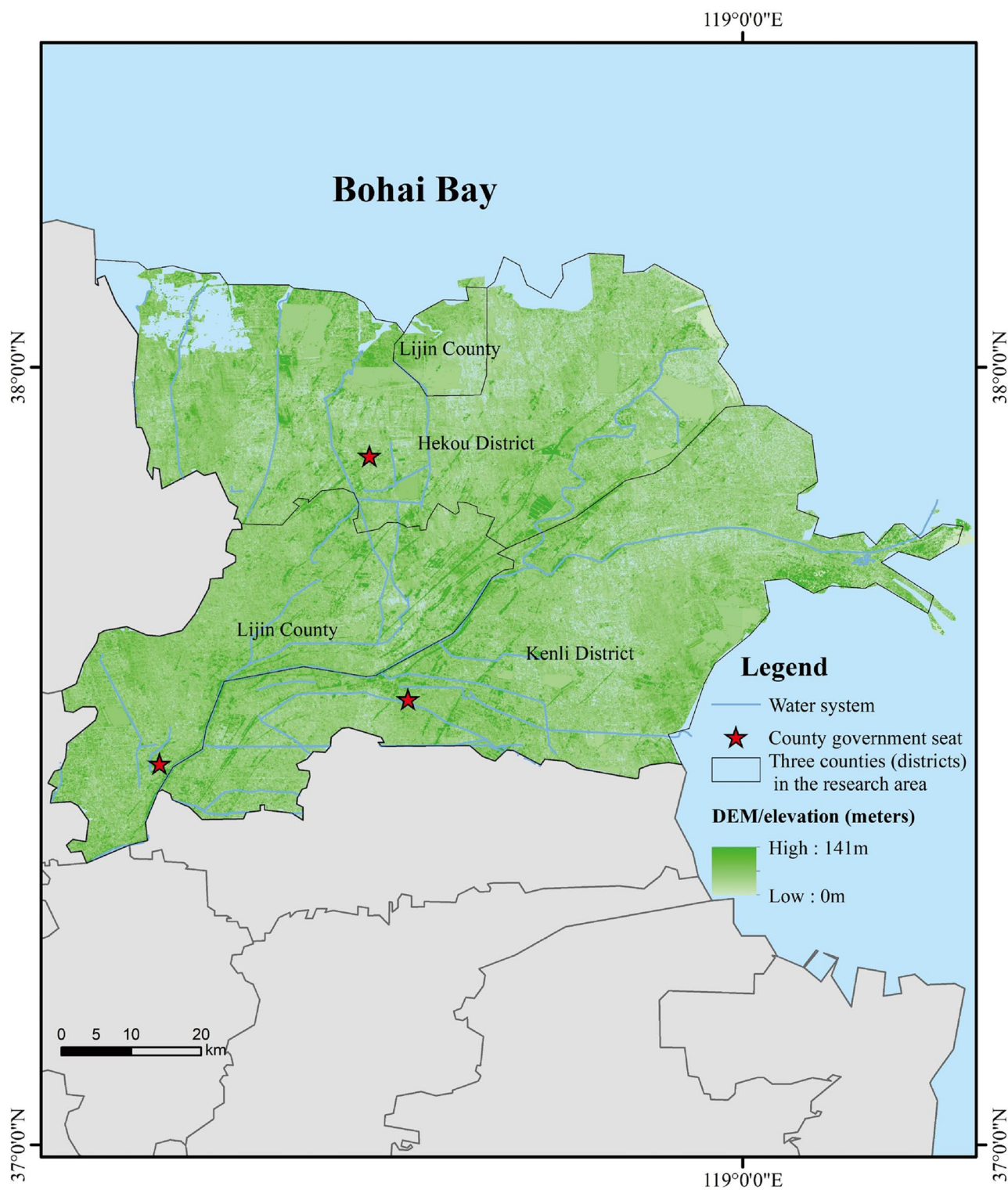


Fig. 2 Overview of the study area. Map lines delineate study areas and do not necessarily depict accepted national boundaries, the same below.

Table 1 Data sources.

data type	Data name	data sources
Basic data	Land use data	Wuhan University (https://zenodo.org)
Socio-economic factors	Population density	Resource and environmental science and data center, Chinese Academy of Sciences (https://www.resdc.cn/)
	GDP	
Distance factor	Distance from class I road	National Geographic Information Resources Directory Service System (https://www.webmap.cn/)
	Distance from class II road	
	Distance from class III road	
	Distance from water system	
	Distance from county government	
Natural factor	Soil type	National Bureau of Statistics (https://www.stats.gov.cn/)
	Annual average temperature	Resource and environmental science and data center, Chinese Academy of Sciences (https://www.resdc.cn/)
	Annual precipitation	
	DEM	Geospatial data cloud (https://www.gscloud.cn/search)
	Slope	Generated by DEM

selected based on the impact of the natural environment and social economy on landscape in the study area⁴⁵. In addition, the study unified the data projection coordinates as WGS 1984 UTM Zone 50 N, resampled the raster data with a spatial resolution of 30 m × 30 m, and kept the number of rows and columns consistent.

Research methods

PLUS model

The PLUS model is a land use simulation tool based on cellular automata. It not only has high simulation accuracy, but also has the ability to measure landscape patterns more realistically. It includes two modules: Land Expansion Analysis Strategy (LEAS) and CA-based Randomized Seed (CARS)^{53,54}. For future simulations, the land use demand projected for 2033 is derived from the observed transition patterns between 2013 and 2023, and is integrated into the PLUS model as a quantitative constraint.

LEAS

$$P_{i,j}^d(x) = \frac{\sum_{n=1}^M I[h_n(x)=d]}{M} \quad (1)$$

In the formula (1), $P_{i,j}^d(x)$ is the probability that sample x is judged as category d_i under the background of model j ; x is the vector composed of driving factors; M is the number of decision trees; $h_n(x)$ is the prediction of sample x by the n th base classifier; $I(\cdot)$ is the indicator function of the decision tree.

CARS

$$OP_{i,j}^{d-1,t} = R_{i,j}^{d-1} \times \Omega_{i,j}^t \times D_j^t \quad (2)$$

In the formula (2), $OP_{i,j}^{d-1,t}$ is the comprehensive probability that space i is transformed into land type j at time t ; $R_{i,j}^{d-1}$ is the basic probability of land type transformation; $\Omega_{i,j}^t$ is the domain weight of land class j in space i ; D_j^t is the adaptive driving coefficient, which affects the future prediction of land type j . The domain weights were determined according to the historical expansion characteristics of different land-use types in the study area and

Table 2 Domain weighting settings.

Land type	Cropland	Grassland	Water	Barren	Built-up area
Domain weight	0.01	0.57	0.8	0.5	1

Table 3 Over-transposed matrix.

Land use type	Cropland	Grassland	Water	Barren	Built-up area
Cropland	1	1	1	1	1
Grassland	1	1	1	1	1
Water	0	0	1	0	0
Barren	1	1	1	1	1
Built-up area	0	0	0	0	1

adjusted during model calibration, with larger values indicating stronger neighborhood expansion capacity. The land-use transition matrix was established based on the observed historical conversion characteristics and actual land-use constraints of the study area (Tables 2 and 3). In the transition matrix, “1” indicates that the corresponding land-use conversion is permitted, whereas “0” indicates that the conversion is restricted⁵⁵.

InVEST model

The habitat quality module in the InVEST model has significant advantages in spatial analysis and accuracy, and is often used to quantitatively assess the habitat quality level in the study area^{46,47}. Based on the land use data, the degree of habitat degradation and habitat quality are calculated according to the Maximum coercive distance and weight of the threat factor, the habitat suitability of different land use types and the sensitivity to threat source^{48,49}. The index is between 0 and 1, and the closer it is to 1, the better the quality of habitat quality⁵⁰.

The formula for Degree of habitat degradation is:

$$D_{xj} = \sum_{r=1}^R \left(\frac{W_r}{\sum_{r=1}^R W_r} - \sum_{y=1}^Y \left(\frac{I_{ry} - S_{jr} - \beta_x}{(d_{xy}/d_{rmax})^2 + 1} \right) \right) \quad (3)$$

In the formula (3), D_{xj} is the comprehensive impact of all threat factors on land use type j in grid cell x ; r is the number of threat factors; R is the total number of threat factors; W_r is the weight of threat factor r ; y is the grid cell of the threat factor r distribution; Y is the number of distribution units of all threat factors; I_{ry} is the threat intensity value of the unit if there is a threat factor r in the unit y ; S_{jr} is the sensitivity of land use type j to threat factor r ; β_x is the Management coefficient; D_{xy} is the distance from grid x to threat factor grid y ; d_{rmax} is the Maximum operating distance of the threat factor r .

Habitat quality formula is:

$$Q_{xj} = H_j \times \left(1 - \frac{D_{xj}^z}{D_{xj}^z + k^z} \right) \quad (4)$$

In the formula (4), Q_{xj} is the habitat quality index of grid x in land type j , with a value between “0” and “1”; H_j is the habitat suitability of land use type j ; z is the default constant of the model; k is the semi-saturated constant, usually taken as 0.5.

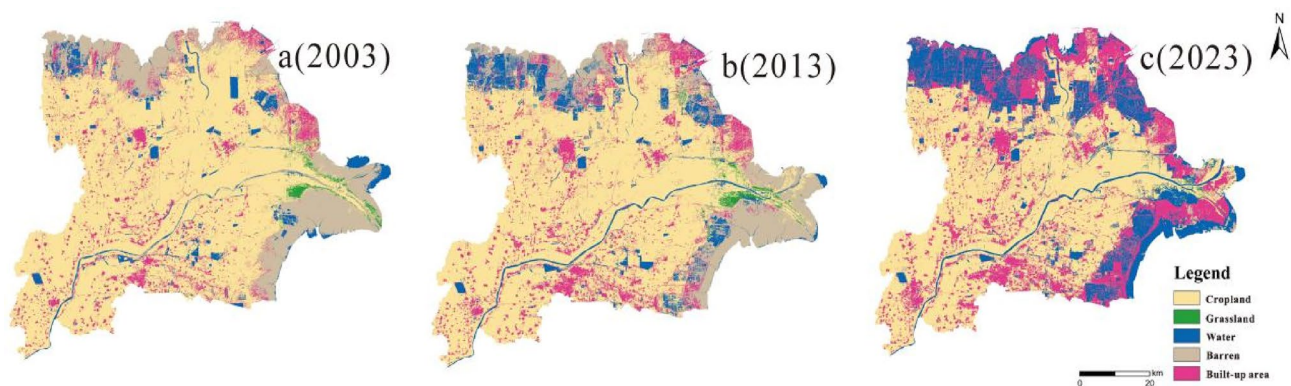
The threat factors and associated parameter values were determined with reference to the InVEST user manual and parameter settings reported in previous habitat quality studies^{51,52}. Particular attention was given to studies conducted in the Yellow River Delta and comparable estuarine environments to improve the regional relevance of the parameterization. The specific parameter sources are provided in Tables 4 and 5.

Table 4 Source of threat attributes.

Threat factor	Maximum influence distance	Relative weight	Distance attenuation type
Cropland	1	0.2	linear
Barren	6	0.5	linear
Built-up area	7	0.6	index

Table 5 Sensitivity table.

Land use type	Habitat suitability	Coercive factors		
		Cropland	Barren	Built-up area
Cropland	0.4	0.3	0.4	0.4
Grassland	1	0.7	0.7	0.6
Water	1	0.7	0.6	0.8
Barren	0.3	0.3	0.1	0
Built-up area	0	0	0	0


Fig. 3 Distribution of land use types in Yellow River Estuary National Park from 2003 to 2023.

Land use transition matrix

The land transition matrix is used to quantify and analyze the transformation relationship between land use types in different periods^{56,57} and reveal the dynamic law of land use change.

$$S_{ij} = \begin{bmatrix} S_{11} & S_{12} & \cdots & S_{1n} \\ S_{21} & S_{22} & \cdots & S_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ S_{n1} & S_{n2} & \cdots & S_{nn} \end{bmatrix} \quad (5)$$

In the formula (5), n is the number of land use types; i, j is the land type before and after the research transfer; S_{ij} represents the area transferred from land type i to land type j .

Results and analysis

Spatio-temporal evaluation of land use and cover change in Yellow River Estuary National Park from 2003 to 2023

From 2003 to 2023, the land use pattern of Yellow River Estuary National Park changed significantly, showing a trend of dual transformation from cropland to construction land and water (Fig. 3). Land use change shows

obvious stage and direction characteristics in spatial distribution characteristics, structural composition and type conversion.

From the perspective of spatial pattern characteristics, cropland is mainly distributed in the central and southern plains of the study area. During the study period, its spatial scope continued to shrink and gradually concentrated in the inland area; The construction land expanded significantly along the traffic trunk lines, urban nodes and the northern and eastern coastal zones, gradually forming agglomeration patches; The water area is concentrated in the main stream of the Yellow River, estuaries and coastal wetland areas, and its spatial scope is significantly expanded, reflecting the continuous implementation of estuarine wetland restoration and water system management projects. Barren is mainly distributed in estuarine sediment areas and coastal beaches, and its spatial scope is significantly reduced, which is closely related to the evolution of coastline evolution and wetland reconstruction process; The grassland is scattered, the overall scale is small, and the change of spatial pattern is not obvious.

From the change characteristics of land use Structural change characteristics (Fig. 4), the area composition and proportion of various land use types have been systematically adjusted during the study period. Cropland and barren continued to decrease, construction land and water area increased significantly, and the land use structure gradually evolved from being dominated by agricultural land to a composite pattern of construction land, water area and agricultural land. This change process reflects that, while urbanization continues to advance, ecological protection and wetland restoration measures have had an important impact on the regional land use pattern.

Based on the land use transition matrix from 2003 to 2023 (Table 6), the internal mechanism of land use structure adjustment is further revealed, and the transformation paths between different land use types are analyzed. The results showed that the land use change in the study area showed obvious directional transfer characteristics, cropland and barren were the main transfer out types, and the transformation process had significant stage differences. From 2003 to 2013, the conversion of cropland to construction land and water conversion was 154.55 km² and 98.30 km² respectively; From 2013 to 2023, the Conversion scale has expanded significantly, and the transformation area from cropland to construction land and Water conversion has increased to 378.11 km² and 225.25

Fig. 4 Proportion of land use types in Yellow River Estuary National Park from 2003 to 2023.

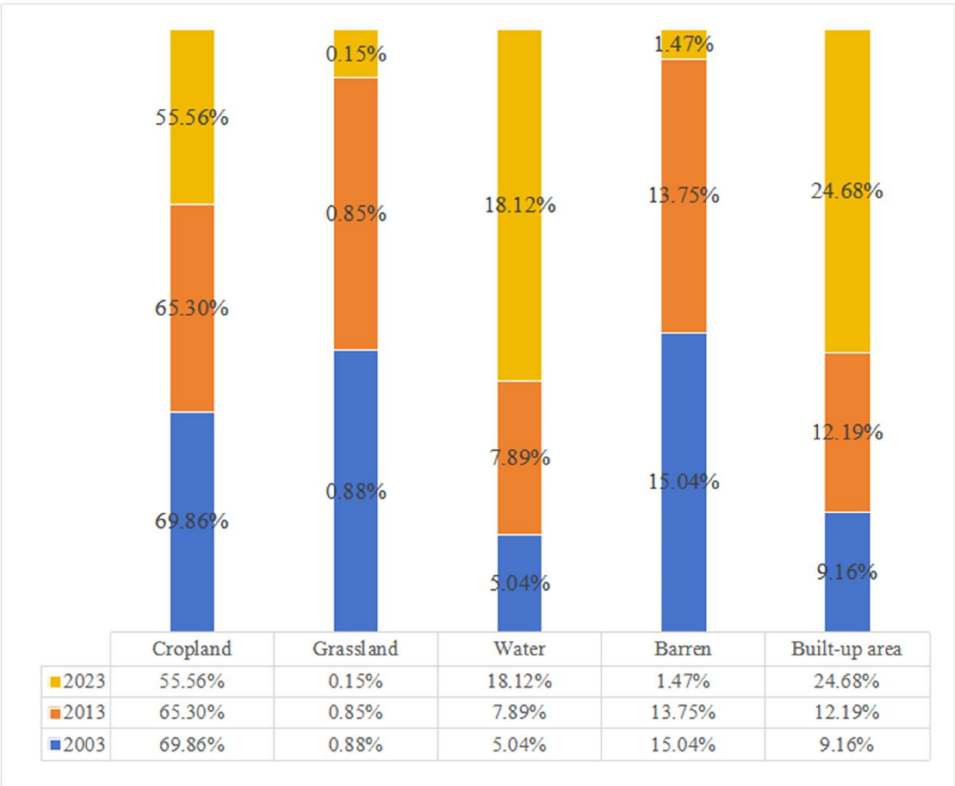


Table 6 Yellow River Estuary National Park land use transfer matrix from 2003 to 2023 (km²).

Land use type		2013				
		Cropland	Grassland	Water	Barren	Built-up area
2003	Cropland	3417.02	26.87	98.30	16.95	154.55
	Grassland	13.61	16.07	2.61	14.35	0.55
	Water	22.28	0.94	175.25	66.76	2.94
	Barren	18.49	1.59	117.57	631.31	30.59
	Built-up area	0.16	0.0027	25.97	1.55	459.37
Land use type		2023				
		Cropland	Grassland	Water	Barren	Built-up area
2013	Cropland	2864.66	2.99	225.25	0.55	378.11
	Grassland	25.38	4.10	10.31	0.54	5.13
	Water	22.39	0.03	356.11	5.77	35.40
	Barren	40.62	1.07	324.54	71.35	293.34
	Built-up area	0.42	0	47.45	0.17	599.96

km², respectively, indicating that the squeezing effect of urban expansion and reconstruction of aquatic ecological space on agricultural land continues to increase. The transformation of barren also showed significant stage characteristics, from 117.57 km² and 30.59 km² in the previous stage to water and construction land, increased to 324.54 km² and 293.34 km², respectively, indicating that the estuarine wetland restoration project and Shore-line evolution had a more prominent impact on the land use pattern in the past decade. In contrast, the grassland conversion scale has always been small, and no dominant transfer path has been formed in either stage, and the overall situation remains relatively stable.

In general, the land use change of Yellow River Estuary National Park from 2003 to 2023 is spatially manifested as the agglomeration expansion of coastal and urban surrounding areas and the continuous retreat of inland agricultural space. Structurally, it is reflected in the significant increase of construction land and water area, and the continuous reduction of cropland and barren. The land use transformation process was formed under the joint action of the development needs of urbanization and the national ecological protection policy, which provided a spatial basis for the subsequent analysis of the evolution characteristics of habitat quality.

Future land use prediction based on PLUS model

PLUS model validation

To evaluate the predictive performance of the PLUS model, the land-use changes observed from 2003 to 2013 were used to simulate the land-use pattern in 2023. The simulated 2023 land-use map was then compared with the observed 2023 land-use map to assess the agreement between predicted and actual spatial patterns (Fig. 5). Overall accuracy and the Kappa coefficient were used as validation metrics. The results showed an overall accuracy of 0.85 and a Kappa coefficient of 0.7567 (>0.75), indicating an acceptable level of agreement between the simulated and observed land-use patterns^{58,59}. Based on this validation, the land-use transition characteristics observed during 2013–2023 were subsequently used for the 2033 scenario simulation.

Analysis on driving factors of land use expansion

In order to explore the spatial pattern and its driving mechanism of land use change in Yellow River Estuary National Park, the study analyzed the land use data of Yellow River Estuary National Park from 2003 to 2023 by using the Random Forest Algorithm and the LEAS module, and evaluated the contribution of each driving factor in different land use changes (Fig. 6). The results show that the land use pattern in the study area is mainly driven

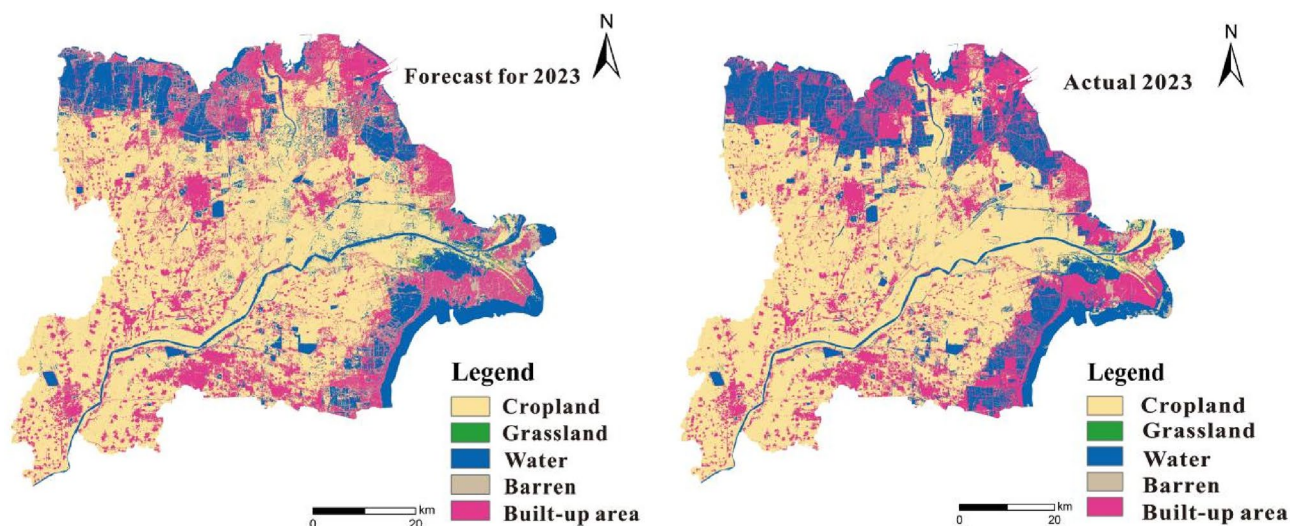


Fig. 5 Comparison between prediction and practice of Yellow River Estuary National Park in 2023.

by the radiation effect of the Administrative center radiation effect and the development of the transportation network, and its formation is closely related to the promotion of urbanization and industrial development.

The spatial distribution of cropland, grassland and barren showed a marginal gradient with the county government as the core and increasing around, and its area was negatively correlated with the distance from the city center. This pattern reflects the spatial process of the continuous agglomeration of population, economy and public resources to the administrative center in the process of urbanization. As an area with a high concentration of population and economic activities, the land use intensity around the administrative center has been continuously improved, and a large number of agricultural and ecological land has been converted into construction land, resulting in the significant compression of cropland, grassland and barren in the core area of the town. With the increase of the distance from the town, the intensity of human activities has gradually weakened, and can be relatively concentrated in the peripheral area. This land use development pattern, dominated by the expansion of construction land, not only improves the carrying capacity of cities and towns, but also weakens the spatial integrity of agricultural and ecological land to a certain extent. From the perspective of socioeconomic background, China's urbanization process has long presented a rapid development mode characterized by spatial expansion and factor agglomeration, which not only promotes regional economic vitality, but also forms a continuous squeeze on agriculture and ecological space.

On the basis of administrative center radiation effect, traffic conditions further magnified the impact of urbanization and industrial activities on land use patterns. As an important infrastructure for animal husbandry and industrial development, the layout of the class II roads has significantly improved regional accessibility and efficiency of factor flow in production, directly supporting the belt and node expansion of construction land. By integrating the regional internal transportation system, it promotes the agglomeration and development of industries along the line that rely on traffic conditions, such as animal husbandry and salt industry, and thus has a far-reaching impact on the spatial layout of water and related productive land. The population agglomeration effect of the administrative center and the industrial amplification effect of traffic conditions are superimposed, which jointly promote the transformation of the land use structure in the study area to artificialization and high-intensity utilization, and continue to squeeze the low-intensity use of cropland, grassland and barren.

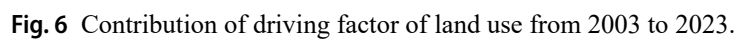


Fig. 7 Land use prediction of Yellow River Estuary National Park in 2033.

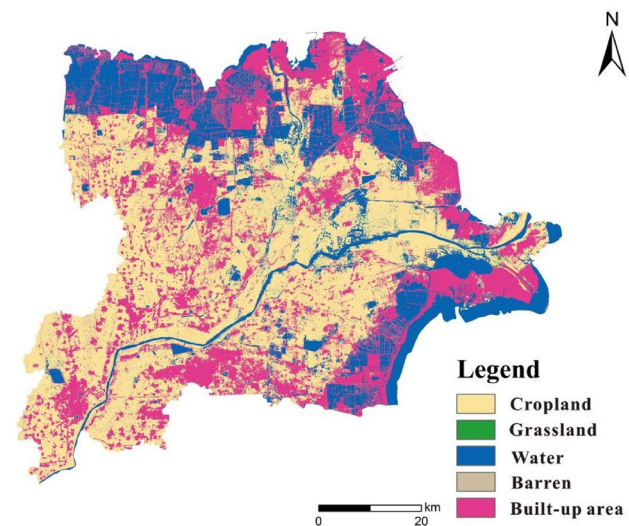


Table 7 Land use transition matrix from 2022 to 2033 (km²).

Land use type		2033				
		Cropland	Grassland	Water	Barren	Built-up area
2023	Cropland	2495.76	0.30	133.50	0.0054	323.91
	Grassland	0.93	3.01	3.83	0.0036	0.41
	Water	0	0	963.69	0	0
	Barren	1.66	0.15	26.83	35.99	13.73
	Built-up area	0	0	0	0	1311.97

Land use prediction of Yellow River Estuary National Park in 2033

The prediction results of land use pattern of Yellow River Estuary National Park in 2033 are shown in Fig. 7. The area of each land use type is 2498.36 km² of cropland, 3.48 km² of grassland, 1127.87 km² of water area, 36.00 km² of barren and 1650.04 km² of construction land. Cropland is still the largest land use type in the study area, but its proportion has decreased to 46.99%, indicating that the trend of agricultural land shrinking will continue; The proportion of grassland and barren decreased synchronously, indicating that it will tend to be marginalized in the future land cover pattern. In contrast, the water area will increase to 1127.87 km² (21.21%), reflecting the sustainable role of wetland restoration and water ecological protection measures in the future scenario; construction land maintained a significant expansion trend, accounting for 31.04%, becoming the most prominent type of land use structure change.

Combined with the analysis of land use transition matrix in 2033 (Table 7), the land use change in the study area shows the multipath flow characteristics with Cropland conversion as the core. While maintaining the core distribution area of 2495.76 km², cropland is still under the dual transformation pressure of construction land (323.91 km²) and water area (133.50 km²), reflecting the compound impact of urbanization expansion and ecological restoration projects on agricultural space; The transformation of grassland (3.83 km²) and barren (26.83 km²) to water shows that the expansion trend of wetland ecosystem is still continuing, while the transformation of barren to construction land (13.73 km²) shows that it is still under certain development pressure in some areas.

On the whole, the expansion of construction land is characterized by path dependence with cropland conversion as the main factor, and the growth of water area is mainly from cropland and barren conversion, which reveals the characteristics of multiple circulation and structural differentiation of the land use pattern of Yellow River Estuary National Park under the dual drive of ecological protection policy and regional economic development.

Change and simulation of habitat quality changes in Yellow River Estuary National Park based on InVEST model

Spatio-temporal evolution characteristics of habitat quality from 2003 to 2023

Based on the land use data of Yellow River Estuary National Park from 2003 to 2023, the Habitat Quality module of the InVEST model was used to calculate changes in habitat quality. According to the natural breaks method (Jenks) in ArcGIS 10.8 and referring to relevant literature^{60,61}, habitat quality in the study area was classified into five levels: low [0, 0.2), moderate-low [0.2, 0.4), moderate [0.4, 0.6), moderate-high [0.6, 0.8), and high [0.8, 1.0]. To avoid interval overlap, the half-open interval principle was adopted to ensure that each value has a unique attribution.

From the perspective of spatial pattern (Fig. 8), the habitat quality of Yellow River Estuary National Park from 2003 to 2023 showed a significant differentiation characteristic of “high in coastal areas and low in inland areas”. Moderate-high and high habitat quality areas are mainly concentrated in the Yellow River Estuary and coastal wetland areas, which are dominated by natural water bodies and wetland ecosystems with relatively weak human disturbance; while low and moderate-low habitat quality areas are widely distributed in inland cropland areas and urban built-up area agglomeration areas, and highly coincide with high-intensity agricultural utilization, urban expansion, and transportation infrastructure layout, indicating that land use mode is the key factor driving the spatial differentiation of habitat quality.

From the perspective of structural change characteristics (Fig. 9), the composition of habitat quality levels has been significantly adjusted during the study period. The proportion of low habitat quality continued to increase, becoming the most dominant habitat quality type in the study area; moderate-low habitat quality decreased significantly, while moderate habitat quality increased in the late stage. Moderate-high and high habitat quality showed restorative growth after reaching a phased trough in 2013, but their spatial distribution tended to be fragmented, patch sizes were small, and overall connectivity remained weak. In general, the structure of habitat quality levels in the study area shows a complex pattern of low-quality expansion and high-quality local restoration.

In order to further reveal the internal mechanism of habitat quality evolution, the analysis is based on the transition matrix of habitat quality from 2003 to 2023 (Table 8). The results showed that habitat quality changes had obvious directional and stage-based characteristics. From 2003 to 2013, the degradation of habitat quality was mainly from moderate-low to low habitat quality; from 2013 to 2023, the degradation trend was further intensified, and the transformation area from moderate-low to low habitat quality reached 702.45 km². At the same time, the transformation from high to moderate-low habitat quality existed in both stages, indicating that high-quality habitat is still facing continuous human activity pressure. In terms of habitat quality improvement, the improvement process mainly occurred within moderate-low habitat quality. During 2013–2023, the transformation areas from

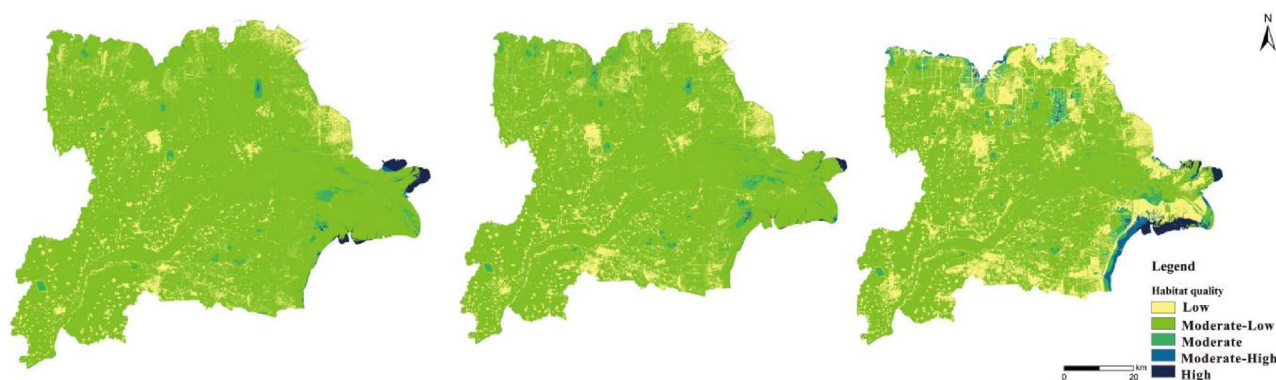


Fig. 8 Spatial distribution of habitat quality in Yellow River Estuary National Park from 2003 to 2023.

Fig. 9 Proportion of Habitat quality level of Yellow River Estuary National Park from 2003 to 2023.

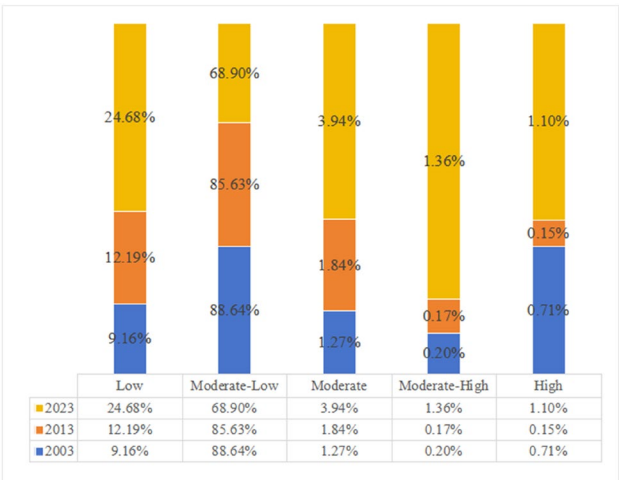


Table 8 Transition matrix of habitat quality of Yellow River Estuary National Park from 2003 to 2023 (km²).

Habitat quality level		2013				
		Low	Moderate-low	Moderate	Moderate-high	High
2003	Low	459.37	27.65	0.04	0	0
	Moderate-low	188.24	4464.56	52.01	6.003	1.29
	Moderate	0.38	21.17	46.21	0.0054	0
	Moderate-High	0.0081	7.63	0.0009	3.12	0
	High	0.0054	30.9	0	0	7.08
Habitat quality level		2023				
		Low	Moderate-low	Moderate	Moderate-high	High
2013	Low	599.96	44.56	3.37	0.12	0
	Moderate-low	702.45	3589.36	143.24	65.96	50.92
	Moderate	8.73	26.65	62.89	0.0009	0
	Moderate-high	0.78	1.68	0.0009	6.66	0
	High	0.036	0.63	0	0	7.70

moderate-low to moderate-high and high habitat quality were 65.96 km² and 50.92 km², respectively, reflecting that estuarine wetland protection and ecological restoration measures have achieved certain results in some areas.

Overall, habitat quality evolution in Yellow River Estuary National Park from 2003 to 2023 showed general characteristics of degradation and local improvement. Low habitat quality continues to expand spatially, while moderate-high and high habitat quality have recovered, but due to patch fragmentation and insufficient connectivity, it is difficult to form a stable expansion pattern. The evolutionary characteristics of habitat quality provide a realistic background for subsequent scenario simulation analysis and further highlight the dominant role of land use change in habitat quality.

Habitat quality simulation of Yellow River Estuary National Park in 2033

Based on the simulation results of land use in 2033, the InVEST model was used to predict the future habitat quality pattern of Yellow River Estuary National Park (Fig. 10; Table 9). The results show that future habitat quality will exhibit an evolutionary pattern dominated by low-quality expansion, and the overall trend significantly deviates from the improvement goal emphasized by high-quality ecological environment development. Spatially, low habitat quality continues to expand and form clusters in and around urban agglomeration areas, while moderate-low habitat quality continues to be replaced by it in the western and central regions, reflecting the continuous squeezing effect of built-up area expansion and human activity intensity on habitat quality. At the

Fig. 10 Prediction of habitat quality in 2033.

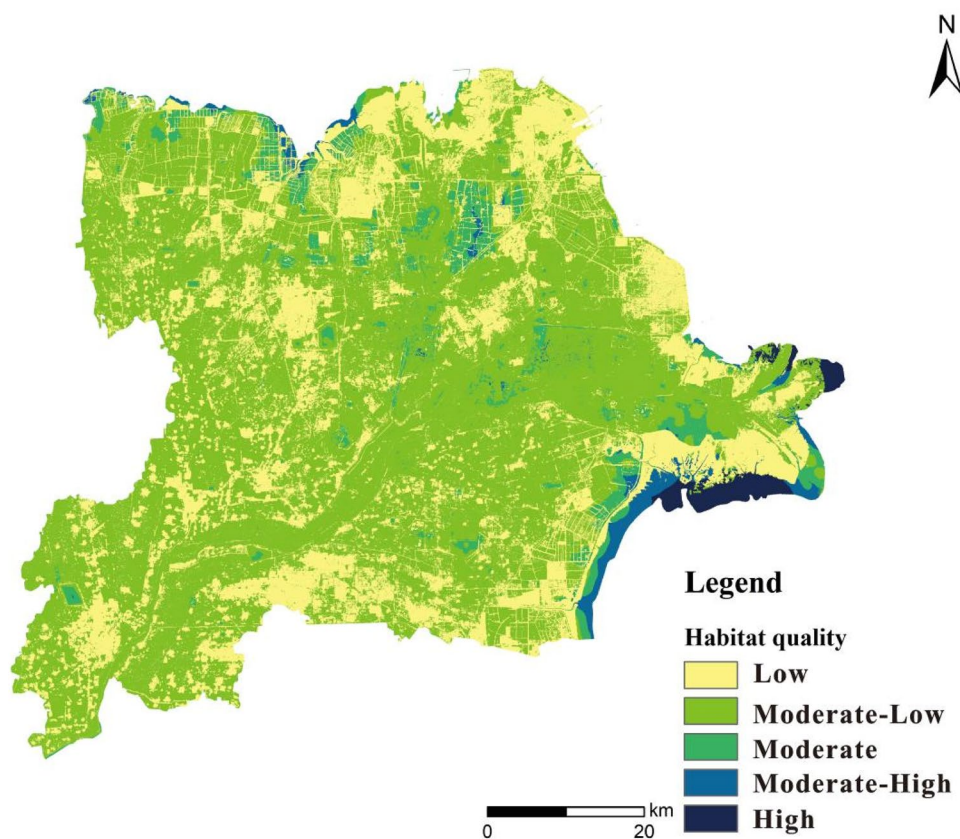


Table 9 Transition matrix of habitat quality from 2023 to 2033/km².

Habitat quality level		2033				
		Low	Moderate-low	Moderate	Moderate-high	High
2023	Low	1311.97	0	0	0	0
	Moderate-low	337.75	3257.06	54.94	7.91	5.21
	Moderate	0.27	1.37	207.86	0	0
	Moderate-high	0.03	1.68	0.0018	72.46	0.02
	High	0.0045	0.13	0	0	58.49

same time, although high habitat quality is limited to core wetland areas, patch sizes have shrunk, fragmentation has intensified, and spatial connectivity has declined. Moderate habitat quality is scattered in coastal zones and around river channels, with limited transition and buffer functions, making it difficult to support the expansion and agglomeration of high-quality habitat.

The characteristics of habitat quality transfer further reveal internal changes in spatial pattern formation. From 2023 to 2033, the maintenance area of low habitat quality reached 1131.97 km², with almost no transformation to higher habitat quality levels, indicating that, under current land use structures and human activities, once low-quality habitat is formed, it tends to persist for long periods due to built-up area expansion, ecological space destruction, and limited natural restoration conditions. Although moderate-low habitat quality remains the dominant type (3257.06 km²), 337.75 km² has transformed into low habitat quality, indicating that habitat structure and function in regions with certain ecological functions have been continuously weakened under sustainable agricultural utilization and urbanization, ultimately leading to quality degradation. In contrast, although moderate-high and high habitat quality maintained areas of 72.46 km² and 58.49 km², respectively, there was still transformation to lower-grade habitat quality, indicating that under conditions of high-intensity human activities and hindered

spatial continuity, it is difficult for high-quality habitats to achieve stable expansion. In general, future habitat quality evolution is subject to the expansion of low habitat quality and the fragmentation and transformation of high-quality habitats, which together constitute the core structural factors restricting overall regional habitat quality improvement.

Discussion

The observed evolution of habitat quality has direct implications for land quality and ecosystem services in the Yellow River Estuary National Park⁶². The expansion of low-quality habitat associated with built-up land growth and intensive land use indicates a decline in the landscape's capacity to sustain biodiversity and related regulating services. In contrast, the localized recovery of relatively high-quality habitat in wetland and water-dominated areas suggests that ecological restoration can improve habitat conditions in specific locations. However, these improvements remain spatially limited and are insufficient to fully offset the broader pressures generated by continued land development. The coexistence of degradation and localized recovery indicates that habitat quality evolution in the Yellow River Estuary should not be interpreted as a uniform process, but rather as the combined outcome of different land-use transitions and their spatially differentiated ecological consequences.

This interpretation is broadly consistent with previous studies showing that urban expansion, agricultural intensification, and transportation infrastructure development are important drivers of habitat degradation in estuarine regions^{63,64}. However, the present study further indicates that the ecological consequences of land-use change vary with both the direction of land conversion and its spatial context. The expansion and concentration of built-up land intensify landscape fragmentation and reinforce the persistence of low-quality habitat, particularly in inland and peri-urban areas. By contrast, the expansion of water-related ecological space and the restoration of wetland areas contribute to localized improvements in habitat quality. This finding is consistent with previous evidence that ecological restoration can alleviate habitat degradation to some extent, while also suggesting that localized restoration alone is insufficient to offset the broader ecological pressures associated with continued land-use restructuring⁶⁵. Accordingly, the observed pattern of relatively high habitat quality in coastal and estuarine areas and lower habitat quality in inland areas reflects not only differences in natural conditions, but also the spatial interaction among development pressure, land-use change, and ecological restoration.

These spatial differences have important implications for the management of the Yellow River Estuary National Park. Areas characterized by persistently low habitat quality and continued expansion of built-up land should prioritize the control of additional development pressure and the prevention of further fragmentation of ecological space. Coastal wetlands and estuarine areas with relatively high habitat quality should focus on maintaining the integrity and continuity of existing habitat patches while limiting new disturbances. In areas where habitat quality has improved alongside the expansion or restoration of water-related ecological space, management should place greater emphasis on consolidating restoration outcomes and strengthening ecological connectivity with surrounding habitats. Therefore, ecological restoration should be differentiated according to the spatial characteristics of land-use change and habitat quality, rather than relying on uniform interventions across the entire national park. Such spatially targeted management can enhance the practical value of habitat quality assessment for conservation zoning, restoration priority setting, and adaptive ecological management.

Several limitations should nevertheless be acknowledged. First, the InVEST habitat quality model is sensitive to the weights, influence distances, habitat suitability values, and sensitivity coefficients assigned to different threat factors. Although these parameters were derived from previous studies and adjusted according to the characteristics of the study area, they cannot fully capture the complexity of ecosystem responses to anthropogenic disturbance. Second, the PLUS-based projections primarily reflect the continuation of historical land-use transition patterns and do not fully account for uncertainties associated with extreme climate events, hydrological change, or major policy adjustments. Future studies should therefore integrate longer time-series data, field-based

ecological observations, and spatial management units to further evaluate the robustness of habitat quality patterns and translate model-based assessments into more precise restoration priorities and management strategies.

Conclusion

Based on the InVEST-PLUS coupling model, this study constructed a spatiotemporal evolution assessment and simulation prediction system for habitat quality in Yellow River Estuary National Park from the perspective of land use change, and systematically revealed the differentiation characteristics and response mechanisms of habitat quality patterns under different human activities. The results show that land use transformation and its spatial pattern characteristics have a decisive impact on habitat quality evolution.

- (1) Habitat quality in Yellow River Estuary National Park showed significant spatial heterogeneity. Coastal wetlands and estuarine waters are dominated by continuous natural patches, which maintain high habitat quality and show strong ecological stability. However, low habitat quality in inland urban expansion areas and high-intensity agricultural utilization areas continued to aggregate, forming a pattern of habitat degradation dominated by human disturbance. This difference reflects the imbalance between natural ecological space and human land expansion in spatial competition.
- (2) Different land use evolution paths shape different habitat response patterns. The centralized expansion of built-up area aggravates landscape fragmentation by compressing natural habitat space, and significantly reduces regional habitat quality. Ecological patches dominated by wetlands and water bodies have buffered human activity pressure to a certain extent by maintaining landscape connectivity, forming a low-interference, high-quality habitat maintenance mechanism.
- (3) Habitat quality evolution has obvious stage-based and nonlinear characteristics. In the early stage of the study, rapid urbanization advancement continued to suppress habitat quality. In the later stage, with the intervention of ecological protection and restoration measures, local habitat quality was restored, but low habitat quality still showed an expanding trend, indicating that the cumulative effect of human activity intensity remains dominant.
- (4) Scenario simulation reveals the potential risks of future habitat quality evolution. The prediction results show that continuous built-up area expansion in the future will strengthen the stability of low habitat quality. Although high habitat quality patches can maintain a certain scale, their spatial connectivity will decline, and overall ecosystem resilience will face the risk of weakening.
- (5) The study highlights the critical role of habitat quality as a proxy for ecosystem services in the context of land use change. The observed patterns of degradation and recovery have direct consequences for land quality, suggesting that future research should incorporate soil and vegetation attributes to fully assess the multifaceted impacts of human activities on estuarine landscapes. This integrated approach will better support policy-making aimed at balancing development with ecological conservation in national parks and similar protected areas.

The research shows that habitat quality evolution in Yellow River Estuary National Park is essentially the result of the coupling effect of land use transformation and ecosystem response. The InVEST-PLUS coupling model can provide quantitative support and methodological reference for zoning management and control of national parks, identification of priority areas for wetland protection and restoration, and integrated land-sea ecological governance.

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Data availability The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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Authors and Affiliations

Zhuang Zhuang^{1,2} · Zhaojin Li^{1,2} · Pengfei Tai^{1,2} · Fugao Jiang^{1,2}

✉ Pengfei Tai
taipengfei268@qfnu.edu.cn

¹ School of Sports Science, Qufu Normal University, Jining 273165, China

² Shandong Rural Revitalization Sports Research Institute, Qufu Normal University, Jining 273165, China